

Figure 9: Data security

such levels. Cloud providers have data centres in multiple zones, and the hardware (like servers, routers, cables, etc) is physically isolated and secured from any unauthorised access, any natural disaster, and from theft.

The main concern with on-premise infrastructure is related to its ageing and obsolete technology. Cloud service providers use the most advanced, up-to-date technology to ensure against any damage to hardware or stored data.

Virtual security threats

Virtual security can be further divided into two categories.

Network security threats: There are many network related threats. With the advancement of technology, there is a rapid growth in the types of these threats. Every time a solution is found, new threats emerge. These threats clearly depend upon the malicious intent of human beings. Examples of such threats are DoS attacks, phishing attacks, MITM attacks, etc.

All these threats have been identified and mitigated to a large extent. It is not like there is no cure for any emerging malady. Of course, we humans are responsible for every loophole, as we make mistakes. If we think the way the attacker thinks, and if we take care of every small glitch, we can minimise the risks to a great extent. So here are some precautions that need to be taken at the time of set-up. Remember, prevention is always better than cure.

1. Make an access list and provide access only to

swarm to the cloud these days. The cloud not only takes care of the most satisfying feature called availability, i.e., more than 99 per cent, but it also takes care of the reliability and durability of resources to a great extent. On-premise infrastructure can never dream of reaching

those on the list.

2. Maintain a system to identify all unwanted access and activity, and analyse intrusions properly. Do not promote the 'let it go' attitude within the organisation.
3. Use secure protocols for remote login, such as SSH.
4. Use a difficult password.
5. Configure access lists or firewalls to permit only necessary traffic.

New and unique security solutions can be identified depending on your set-up and requirements.

Data security threats: With technology advancing and with the development of cloud computing, data security has turned into a major concern. Rising concerns have led to the development of some important methods to fulfil security requirements. These include advanced authentication encryption, key management, HSM (hardware security module) integration with safe-net and, of course, data masking.

Data security needs to achieve three main goals.

- **Confidentiality** is privacy. To keep data private, it is necessary to maintain the privacy of each form of data. Not only is the privacy of sensitive data of the utmost priority, but the metadata and transactional data also needs to be safeguarded. There are many ways to secure data as described above, such as encryption, access control, any type of data masking as well as logical preventions.
- **Integrity** of data is the non-adulteration of data. No matter what happens, data integrity has to be maintained. Think of the consequences if your surname changes in your voter ID card. Data integrity can be achieved with a little effort. A well audited code also helps in maintaining the integrity of data. Data integrity is of utmost priority in case of revision control systems such as Git or Mercurial. They allow users to roll back previous versions of saved data. These systems use the SHA-1 algorithm to ensure data has not changed because of any accidental corruption.
- **Availability** is the third most important goal. What is the use of that voter ID card if you cannot vote for the upcoming elections? You think of using a bidding application but are prevented from bidding because you cannot access your banking site. Availability can be achieved with widespread Internet enabled access, and this is the topmost aim of any cloud service provider. AWS uses CloudFront to provide low latency in data access.

Managing your cloud

Management is the key while maintaining the security and integrity of your cloud. This has been a very hot topic since the development of cloud technology. There are tools that provision, monitor and can audit the activity on the cloud.

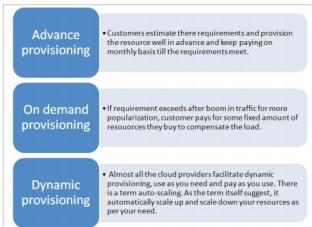


Figure 10: Resource provisioning

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